

The command syntax to display the file 'sample.txt' one page at a time is *

1/1

- ☐ man sample.txt>more
- ☐ cat sample.txt<more
- ☒ cat sample.txt|more
- ☐ None of the mentioned

How do you get help about the command "cp"? *

1/1

- ☐ help cp
- ☒ man cp
- ☐ cd
- ☐ None of the mentioned

A process said to be in _____ state if it was waiting for an event that will never occur *1/1

- ☐ Safe
- ☐ Unsafe
- ☒ Deadlock
- ☐ All

The Hardware mechanism that enables a device to notify the CPU is called *1/1

- _____
- ☐ Polling
 - ☒ Interrupt
 - ☐ System Call
 - ☐ None of the above

The address of a page table in memory is pointed by _____ *

1/1

- ☐ stack pointer
- ☒ page table base register
- ☐ page register
- ☐ program counter

Virtual memory is normally implemented by _____. *

1/1

- ☒ demand paging
- ☐ buses
- ☐ virtualization
- ☐ All of these

CPU fetches the instruction from memory according to the value of _____

*1/1

- ☒ program counter
- ☐ status register
- ☐ instruction register
- ☐ program status word

Which of the following command is used to count the total number of lines, words, and characters contained in a file?

*1/1

- ☒ wc
- ☐ wcount
- ☐ countw
- ☐ None

Smaller page tables are implemented as a set of _____ *

1/1

- ☐ queues
- ☐ stacks
- ☐ counters
- ☒ registers

Which command is used to display all the files including hidden files in your current and its subdirectories ?

*1/1

- ☒ ls -aR
- ☐ ls -a
- ☐ ls -R
- ☐ ls -l

- ☒ allowing a job to use the processor
- ☐ making proper use of processor
- ☐ Both i and ii
- ☐ None of these

Correct answer

- ☒ Both i and ii

Banker's Algorithm is implemented to *

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- ☐ Detect Deadlock
- ☒ Prevent Deadlock
- ☐ Avoid Deadlock
- ☐ All

Correct answer

- ☒ Avoid Deadlock

which of the following commands can be used to change default permissions `0777` for files and directories at the time of creation

- ☐ chmod
 - ☐ chown
 - ☒ umask
 - ☐ chgrp
-

Which of the following is NOT a mechanism for user programs to request services from the operating system?

1/1

- ☐ API
- ☒ System call
- ☐ Library
- ☐ Assembly instruction

- ☐ d
- ☐ r
- ☒ x
- ☐ c

A problem encountered in multitasking when a process is perpetually denied necessary resources is called *1/1

- ☐ deadlock
- ☒ starvation
- ☐ inversion
- ☐ aging

Which of the following memory allocation scheme suffers from External fragmentation?

*1/1

- ☒ Segmentation
 - ☐ Swapping
 - ☐ Pure demand paging
 - ☐ Paging
-

The command which is used to remove files. ▪

1/1

- ☒ rm
- ☐ rmdir
- ☐ cp
- ☐ mv

A program that acts as an interface between process and OS is called ▪

0/1

- ☐ Kernel
- ☒ System call
- ☐ Microkernel
- ☐ Virtual Machine

Correct answer

- ☒ Kernel

which technique was introduced because a single job could not keep both the CPU and the I/O devices busy?

- ☐ Time-sharing
- ☐ Spooling
- ☐ Preemptive scheduling
- ☒ Multiprogramming

Correct answer

- ☒ Time-sharing

Which of the following commands will allow the user to search contents of a file for a particular pattern *1/1

- ☐ touch
- ☒ grep
- ☐ find
- ☐ ls

Consider three process, all arriving at time zero, with total execution time of 10, 20 and 30 units respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of does the CPU remain idle? *1/1

- ☐ 0%
- ☒ 10.6%
- ☐ 30.0%
- ☐ 89.4%

Paging increases the time *

1/1

- ☐ waiting
- ☐ execution
- ☒ context – switch
- ☐ All of these

A major problem with priority scheduling is _____. *

1/1

- ☐ Definite blocking
- ☒ Starvation
- ☐ Low priority
- ☐ None of the above

When a thread needs to wait for an event it will *

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- ☒ Block
 - ☐ Execute
 - ☐ Terminate
 - ☐ Update
-

The systems which allow only one process execution at a time, are called _____

* 0/1

- ☒ uniprogramming systems
- ☐ uniprocessing systems
- ☐ unitasking systems
- ☐ none of the mentioned

Correct answer

- ☒ uniprocessing systems

Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. 0/1

Perform the allocation of processes using Worst Fit Algorithm

- ☐ Process P1 = 357 KB, Process P2 = 210 KB, Process P3 = 468 KB, Process P4 = 491 KB
- ☐ Process P1 = 210 KB, Process P2 = 357 KB, Process P3 = 468 KB, Process P4 = 491 KB
- ☐ Process P1 = 357 KB, Process P2 = 210 KB, Process P3 = 491 KB, Process P4 = 468 KB
- ☒ None

Correct answer

- ☒ Process P1 = 357 KB, Process P2 = 210 KB, Process P3 = 468 KB, Process P4 = 491 KB

Semaphores are used to solve the problem of *

0/1

- ☒ race condition
- ☐ process synchronization
- ☐ mutual exclusion
- ☐ belady problem

Correct answer

- ☒ process synchronization

Semaphore is a/an _____ to solve the critical section problem. *

1/1

- ☐ hardware for a system
- ☐ special program for a system
- ☒ integer variable
- ☐ None of these

Consider the set of 4 processes whose arrival time and burst time are given below- *1/1

If the CPU scheduling policy is Shortest Remaining Time First, calculate the average waiting time and average turn around time.

Process No.	Arrival Time	Burst Time		
		CPU Burst	I/O Burst	CPU Burst
P1	0	3	2	2
P2	0	2	4	1
P3	2	1	3	2
P4	5	2	2	1

- ☒ 9, 4.4
- ☐ 5, 6.7
- ☐ 3.5, 2.9
- ☐ None

The main function of a command interpreter is ▪

1/1

- ☒ to get and execute the next user-specified command
- ☐ to provide an interface between the API and the application program
- ☐ to handle the files in the operating system
- ☐ None

A process can be terminated due to ▪

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- ☐ normal exit
- ☐ fatal error
- ☐ killed by another process
- ☒ All of the above

Which of the following page replacement algorithms suffers from Belady's anomaly?

*1/1

- ☒ FIFO
- ☐ LRU
- ☐ Optimal Page Replacement
- ☐ Both LRU and FIFO